Akoizo Original Practice - 2027 Rotation Form 3

Division C | Astronomy | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) What can power a protostar before sustained fusion?

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

A. A planetary nebula necessarily

B. No energy source

C. Gravitational contraction

D. Chemical oxygen combustion

Answer: _____

2. (2 points) Hotter blackbodies peak at…

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

A. The same wavelength always

B. Zero frequency

C. Shorter wavelength

D. Longer wavelength

Answer: _____

3. (2 points) An isochrone describes…

Unit: Advanced graph and image interpretation.
Worked-case data: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

A. Equal age across modeled masses

B. One star’s spatial orbit

C. Only equal radii

D. A single observed pulse

Answer: _____

4. (2 points) A black-hole horizon is…

Unit: Supernovae, white dwarfs, neutron stars, and black holes.
Worked-case data: Use 3 km per solar mass as a rounded Schwarzschild-radius scale for a 10 Msun nonrotating black hole.

A. A causal boundary

B. A normal rocky surface

C. A visible shell of gas necessarily

D. The edge of every accretion disk

Answer: _____

5. (4 points) Compute the example luminosity.

Unit: Stellar properties, fusion, and the H-R diagram.
Worked-case data: A star has radius 2 Rsun and temperature 2 Tsun. Calculate L/Lsun.

Answer: _____
_____
_____

6. (4 points) Solve the example distance modulus.

Unit: Astrophysics equations and quantitative synthesis.
Worked-case data: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

Answer: _____
_____
_____

7. (4 points) Distinguish emission and reflection nebulae.

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

Answer: _____
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_____

8. (4 points) What does planetary nebula not imply?

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

Answer: _____
_____
_____

9. (4 points) What can a nondetection mean?

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

Answer: _____
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11. (2 points) A synthetic star has parallax 0.003333333 arcseconds. Use d(pc)=1/p to find distance.

Answer: _____
_____
_____

12. (2 points) A standard-candle exercise gives apparent magnitude 13, absolute magnitude 3 and negligible extinction. Use m-M=5log10(d/10 pc). Find d.

Answer: _____
_____
_____

13. (2 points) A star has twice the reference radius and the same temperature. Find its luminosity ratio using L proportional to R^2 T^4.

Answer: _____
_____
_____

14. (2 points) Use Wien's constant 2.90e-3 m K for a 5,800 K star. Find peak wavelength.

Answer: _____
_____
_____

15. (2 points) An artificial galaxy recession speed is 210 km/s. Use H0=70 km/s/Mpc and the nearby linear Hubble law to find distance.

Answer: _____
_____
_____

16. (2 points) A normalized binary exercise has separation 2 AU and period 2 years. Use total mass in solar masses=a^3/P^2. Find total mass.

Answer: _____
_____
_____

17. (2 points) NASA identifies M82 X-2 as an ultraluminous pulsar. What does a coherent rotation-linked pulse train imply about the compact object, and why is luminosity alone insufficient?

Answer: _____
_____
_____

18. (2 points) SN 2014J is a Type Ia supernova in M82. Explain why standardized Type Ia luminosities can constrain distance and why dust must be considered.

Answer: _____
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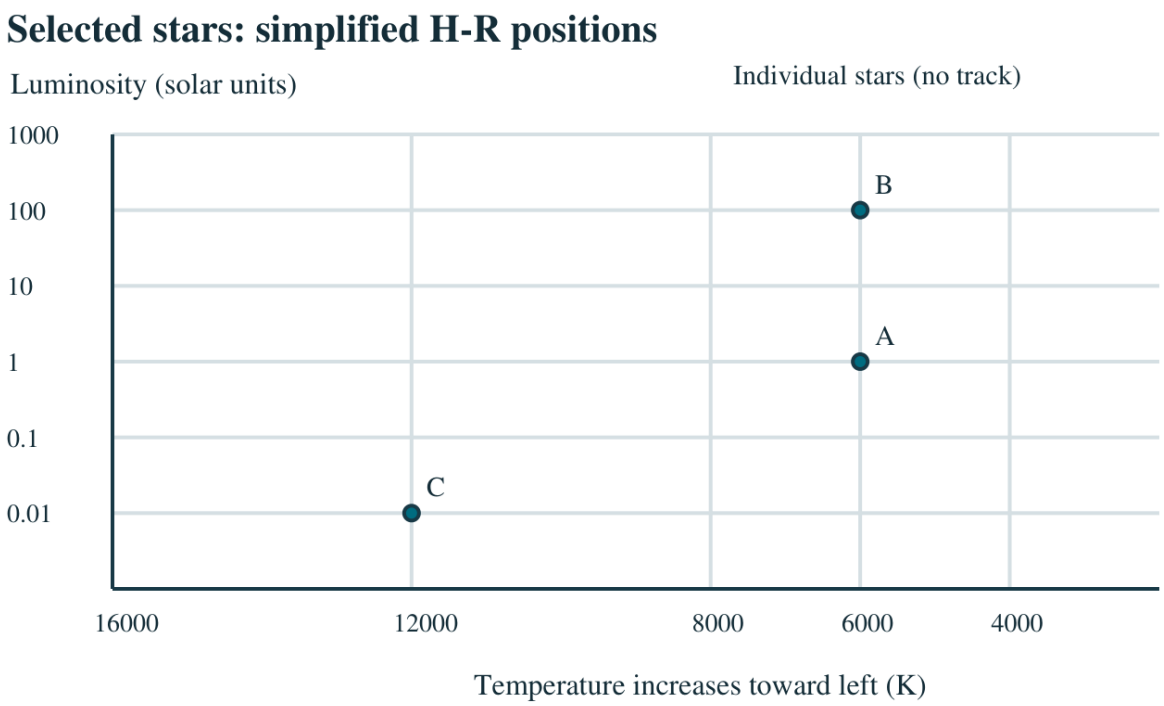
19. (2 points) GW170817 was observed in gravitational waves and electromagnetic radiation. Explain the value of this multi-messenger evidence for a neutron-star merger.

Answer: _____
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20. (2 points) Synthetic starburst-galaxy case 3: ultraviolet emission and infrared dust emission both increase. Explain why using only visible light can miss part of the recent star formation.

Answer: _____
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_____

21. (2 points) Use Figure F1. Use L proportional to R^2 T^4 to find radius B / radius A.



F1. Hertzsprung-Russell diagram (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Which labeled object best fits a hot, low-luminosity compact remnant position? Explain.

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

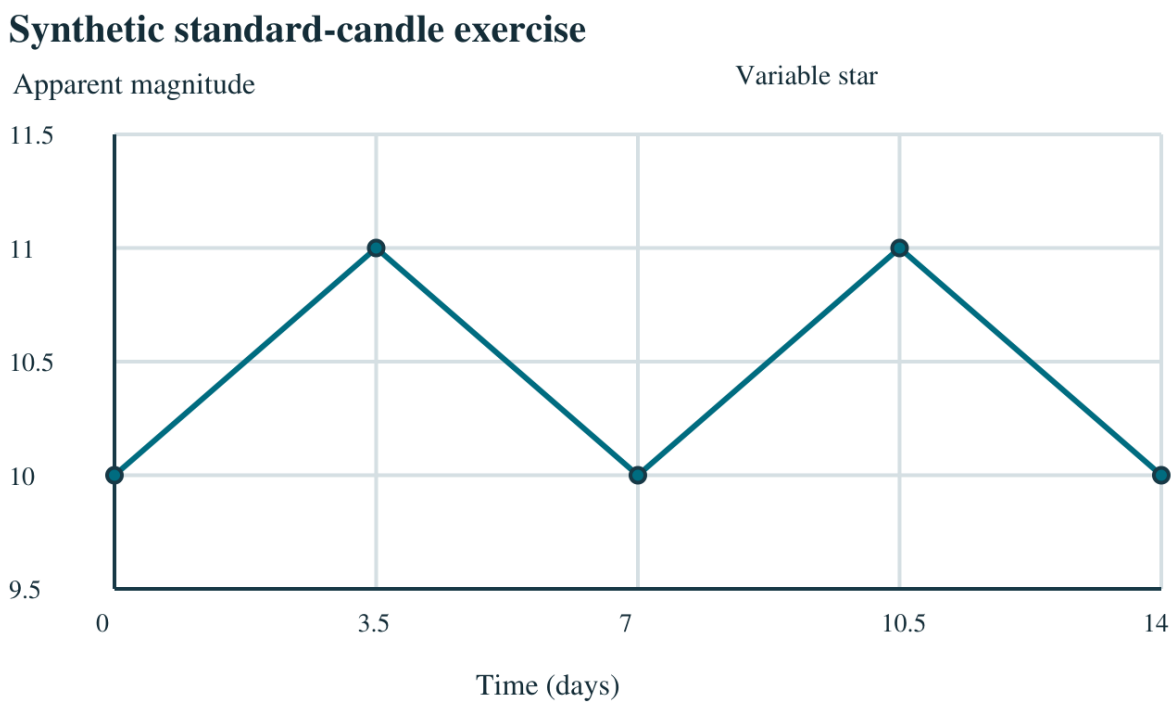
Answer: _____
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23. (2 points) Use Figure F1. Why should the displayed positions not be read as motion of one star through space?

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
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24. (2 points) Use Figure F2. Read the repeat period.



F2. Pulsating-variable photometry (original synthetic teaching figure)

Answer: _____
_____
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25. (2 points) Use Figure F2. For this teaching estimate use the arithmetic midpoint of the plotted magnitude extrema as m. Take M=0.5, negligible extinction, and m-M=5log10(d/10pc). Find distance.

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

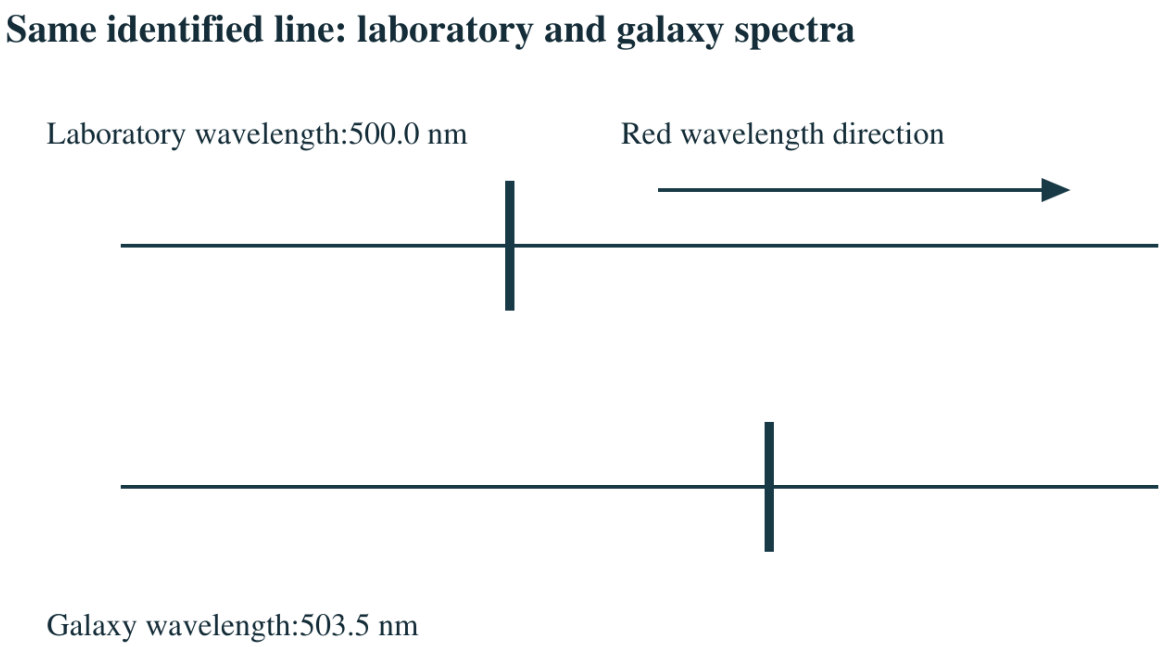
Answer: _____
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26. (2 points) Use Figure F2. At plotted magnitude10 versus11, which state is brighter and why?

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate z=(observed-rest)/rest.



F3. Spectral-line displacement (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Use the low-redshift relation v=cz, with c=300000 km/s, to estimate recession speed.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

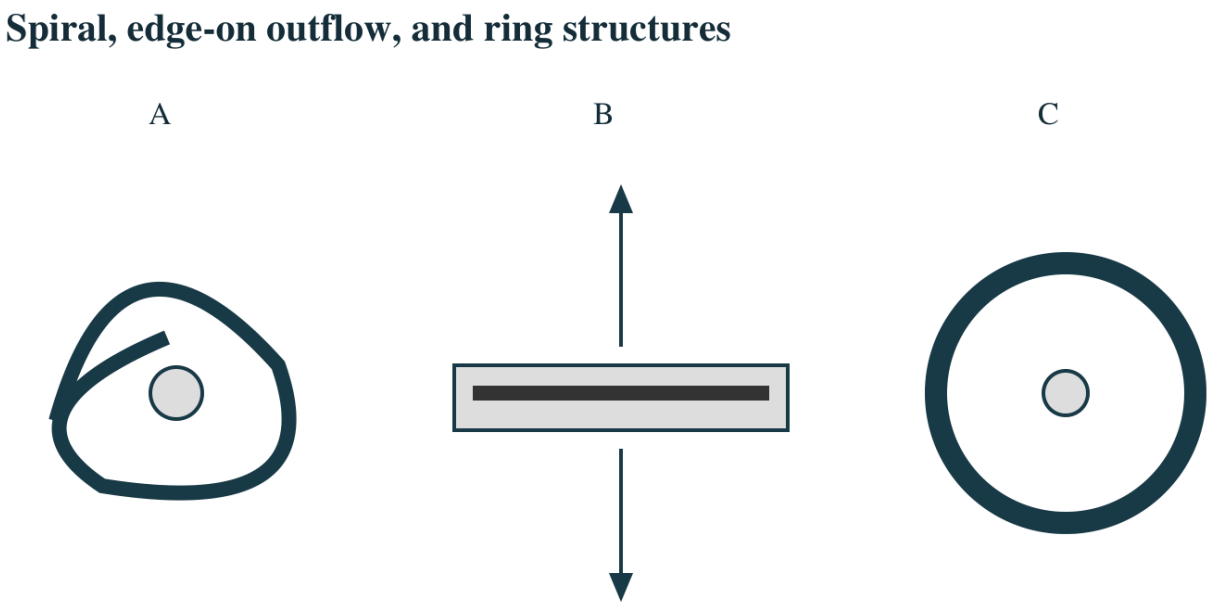
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Use H0=70 km/s/Mpc and the nearby linear Hubble law to estimate distance.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which panel visibly shows a ring morphology, relevant when studying the listed Cartwheel Galaxy?



Original morphology sketches; structures are schematic, not telescope images.

F4. Galaxy morphology comparison (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Which panel illustrates an outflow perpendicular to an edge-on disk, a geometry relevant to starburst-galaxy studies?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
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32. (2 points) Use Figure F4. Why cannot these three morphology sketches alone establish a universal galaxy age sequence?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____